Supplementary File S3 — Search Strategy Documentation

Systematic Review of Hybrid Cyber-Threat Intelligence (CTI) Integration into Healthcare Cybersecurity Risk Assessment (CSRA)

Date of Searches: 25 October 2024

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1. Purpose of This Document

This supplementary file provides a complete and transparent record of the search methods used to identify studies for the systematic review investigating the integration of qualitative, quantitative, and linguistic Cyber-Threat Intelligence (CTI) into healthcare-oriented Cybersecurity Risk Assessment (CSRA) methodologies. The documentation adheres to PRISMA-S guidelines and is supplied to enable full reproducibility of the search strategy, deduplication process, and dataset construction.

2. Review Question

"How can qualitative, quantitative, and linguistic CTI be harmonised into an explainable, auditable, and deployable CSRA methodology suitable for healthcare environments?"

This question guided the selection of databases, controlled vocabulary, and search term combinations.

3. Databases Searched

The following scholarly databases were selected for comprehensive and multidisciplinary coverage:

- Scopus (Elsevier)
- Web of Science – Core Collection (Clarivate)
- IEEE Xplore (IEEE)

- ACM Digital Library (ACM)
- PubMed (MEDLINE)

Searches were performed on 25 October 2024, covering the publication period 1 January 2020 to 25 October 2024.

4. Search Concepts and Rationale

Three conceptual categories informed the search strings:

- **Concept 1: Cyber-Threat Intelligence (CTI)**
  ("cyber threat intelligence", "threat intelligence", CTI)
- **Concept 2: Cybersecurity Risk Assessment (CSRA)**
  ("risk assessment", "security assessment", CSRA)
- **Concept 3: Healthcare context**
  (healthcare, "health care", medical, clinical, hospital)

These concepts were combined using Boolean logic and adapted to database-specific syntax.

5. Database-Specific Search Strategies

5.1 Scopus

(TITLE-ABS-KEY ("cyber threat intelligence" OR "threat intelligence" OR CTI) AND TITLE-ABS-KEY ("risk assessment" OR "security assessment" OR CSRA) AND TITLE-ABS-KEY (healthcare OR "health care" OR medical OR clinical OR hospital) AND PUBYEAR > 2019 AND PUBYEAR < 2025 AND (LIMIT-TO (DOCTYPE, "ar")) OR LIMIT-TO (DOCTYPE, "cp")))

5.2 Web of Science – Core Collection

TS = (("cyber threat intelligence" OR "threat intelligence" O R CTI") AND ("risk assessment" OR "security assessment" O R CSRA") AND (healthcare OR "health care" OR medical O R clinical OR hospital)

Refined by: Document Types: Article OR Proceeding Paper; Publication Years: 2020–2024

5.3 IEEE Xplore

("Abstract": "cyber threat intelligence" OR "Abstract": "threat intelligence" OR "Abstract": CTI) AND ("Abstract": "risk assessment" OR "Abstract": "security assessment" OR "Abstract": CSRA) AND ("Abstract": "healthcare OR "Abstract": "health care" OR "Abstract": medical OR "Abstract": clinical OR "Abstract": hospital)

Filters: 2020–2024; Journals, Magazines, Conference Proceedings

5.4 ACM Digital Library

[Abstract: "cyber threat intelligence"] OR [Abstract: "threat intelligence"] OR [Abstract: CTI] AND [Abstract: "risk assessment"] OR [Abstract: "security assessment"] OR [Abstract: CSRA] AND [Abstract: healthcare] OR [Abstract: "health care"] OR [Abstract: medical] OR [Abstract: clinical] OR [Abstract: hospital]

Filters: Published after 31 December 2019; before 1 November 2024; Research Articles and Conference Papers only

5.5 PubMed (MEDLINE)

("cyber threat intelligence" [Title/Abstract]
OR "threat intelligence" [Title/Abstract]
OR CTI[Title/Abstract])
AND ("risk assessment" [Title/Abstract]
OR "security assessment" [Title/Abstract]
OR CSRA[Title/Abstract])
AND (healthcare[Title/Abstract] OR "health care"[Title/Abstract]

OR medical[Title/Abstract] OR clinical[Title/Abstract]
OR hospital[Title/Abstract])
AND 2020/01/01:2024/10/25[Date - Publication]

Table 1. Complete Database Search Strings and Applied Filters.

Database	Search String / Adaptation	Filters Applied
Scopus	(TITLE-ABS-KEY ("cyber threat intelligence" OR "threat intelligence" OR CTI) AND TITLE-ABS-KEY ("risk assessment" OR "security assessment" OR CSRA) AND TITLE-ABS-KEY (healthcare OR "health care" OR medical OR clinical OR hospital) AND PUBYEAR > 2019 AND PUBYEAR < 2025 AND (LIMIT-TO (DOCTYPE, "ar") OR LIMIT-TO (DOCTYPE, "cp")))	Publication Years: 2020-2024 Document Types: Article, Conference Paper
Web of Science	TS=(("cyber threat intelligence" OR "threat intelligence" OR CTI) AND ("risk assessment" OR "security assessment" OR CSRA) AND (healthcare OR "health care" OR medical OR clinical OR hospital))	Publication Years: 2020-2024 Document Types: Article or Proceeding Paper
IEEE Xplore	("Abstract":"cyber threat intelligence" OR "Abstract":"threat intelligence" OR "Abstract":CTI) AND ("Abstract":"risk assessment" OR "Abstract":"security assessment" OR "Abstract":CSRA) AND ("Abstract":healthcare OR "Abstract":"health care" OR "Abstract":medical OR "Abstract":clinical OR "Abstract":hospital)	2020-2024 Journals & Magazines, Conference
ACM DL	[Abstract: "cyber threat intelligence"] OR [Abstract: "threat intelligence"] OR [Abstract: CTI] AND [Abstract: "risk assessment"] OR [Abstract: "security assessment"] OR [Abstract: CSRA] AND [Abstract: healthcare] OR [Abstract: "health care"] OR [Abstract: medical] OR [Abstract: clinical] OR [Abstract: hospital]	Published after 12/31/2019, before 11/1/2024 Research Article, Conference Paper
PubMed	("cyber threat intelligence"[Title/Abstract] OR "threat intelligence"[Title/Abstract] OR CTI[Title/Abstract]) AND ("risk assessment"[Title/Abstract] OR "security	Publication Date: 2020/01/01 to 2024/10/25

	assessment"[Title/Abstract] OR CSRA[Title/Abstract]) AND (healthcare[Title/Abstract] OR "health care"[Title/Abstract] OR medical[Title/Abstract] OR clinical[Title/Abstract] OR hospital[Title/Abstract]) AND 2020/01/01:2024/10/25[Date - Publication]	

6. Search Results and Initial Screening Counts

Table 2. Search results and initial screening counts across all databases. These counts match exactly the values reported in the PRISMA flow diagram.

Database	Records Identified	Records After Deduplication
Scopus	142	142
Web of Science	118	100
IEEE Xplore	95	80
ACM Digital Library	52	40
PubMed	20	12
Total	**427**	**374**

7. Deduplication Procedure

A structured three-stage deduplication protocol was applied using EndNote X9 with manual verification:

1) **Primary matching**: Exact DOI equivalence after canonicalisation (lower-casing, removal of prefixes, whitespace trimming)
2) **Secondary matching**: Normalised title equality after stripping punctuation, stopwords, and case
3) **Tertiary matching**: Fuzzy title similarity (threshold ≥ 0.92) followed by manual adjudication

Key Implementation Details:

- **Tools:** EndNote X9, Zotero, and reproducible Python script (available upon request)
- **Approach:** Conservative methodology, retaining all potentially unique records

- **Verification:** Independent manual checking of ambiguous cases
- **Result:** A total of 53 duplicate records were identified and removed, resulting in 374 unique records for screening.

8. Screening and Eligibility Criteria

The following criteria guided the screening of the 374 unique records.

Inclusion Criteria:

- Peer-reviewed journal articles or conference proceedings
- Empirical evaluation of CTI/CSRA integration
- Healthcare or critical infrastructure context
- Publication date: 2020-2024
- Q1-ranked venues or peer-reviewed proceedings

Exclusion Criteria:

- Purely theoretical/non-empirical studies
- Non-healthcare domains without transferable insights
- Single-modality CTI without fusion approaches
- Non-English publications

Operationalisation and Best Practice Alignment: The narrative inclusion/exclusion criteria were further operationalised through a scoring rubric (Supplementary Table S1) to ensure consistent, measurable application during screening. This approach follows PRISMA 2020 recommendations for enhancing reproducibility in systematic reviews [Page et al., 2021]. The rubric focused on two key dimensions that directly reflect the review's methodological and domain-specific requirements: (1) Empirical Evaluation & Methodological Rigour, and (2) Healthcare Validation Context. All included studies (n=43) met the minimum threshold score (≥3) upon retrospective application of this rubric.

Screening & Eligibility Outcome:
Of the 374 records screened, 262 were excluded based on title and Abstract. The full text of

112 records was sought and retrieved. After full-text assessment, 69 records were excluded for the following reasons:

- Lacked healthcare validation (n = 31)
- Used non-hybrid CTI fusion (n = 22)
- Non-empirical (n = 16)

  A total of 43 studies were included in the final systematic review.

9. Search Limitations

This search strategy acknowledges several methodological limitations:

- Language restriction to English only
- Exclusion of grey literature and preprints
- Focus on major bibliographic databases
- Potential missed studies in non-indexed journals or regional databases

10. Raw Export Files (Important Notice)

The original database export files (e.g., Scopus RIS, Web of Science TXT, IEEE CSV, ACM BioTeX, PubMed XML) are not redistributed in this supplementary package because:

1. Database licence terms prohibit redistribution,
2. Publisher copyright restrictions apply, and
3. Some metadata fields contain controlled or proprietary formats.

To preserve transparency, placeholder files demonstrating the export formats are included in the directory Raw_Search_Results/, accompanied by instructions on how to request the authentic export files:

- from the corresponding author, or
- via the project repository on OSF (DOI: https://doi.org/10.17605/OSF.IO/3CW8Y).

11. Reproducibility Statement

This search strategy is fully reproducible using the documented syntax on the respective database platforms as of October 2024. Database interfaces and search syntax may evolve over time, but the conceptual approach and methodological rigour remain constant.

12. Provenance and Versioning

- **Document version:** 1.0
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All materials in this package are reproducible, traceable, and consistent with ethical research and publication practices.